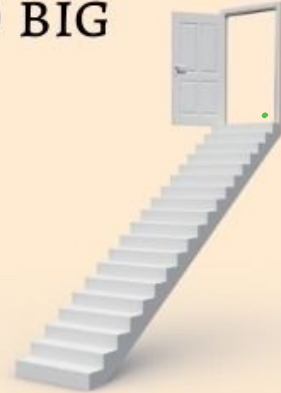


We will
start at
7:05

“A LITTLE PROGRESS EACH
DAY ADDS UP TO BIG
RESULTS.”

SATYA NANI



Introduction

Harshal Garg

-  SDE at Kotak Mahindra Bank | Ex-Amazon | Ex-Oracle
-  Remote Instructor at Scaler for DSA | 175+ Lectures Delivered
-  NIT Jalandhar, B.Tech (CSE) '18
-  Hometown: Agra, UP | Currently residing in Hyderabad

Few terms that you shall see/hear throughout the course!

1. PSP (Problem Solving Percentage) -

Solved Assignment Problems / Total Open Assignment Problems

- There are two types of section - Assignment and Additional. Assignment section consists of implementation of the problems done in class. PSP is calculated based on only Assignment Problems
- Additional Problems are slight modifications of assignment problems, they are not part of PSP but once you're done with assignment, we highly recommend to complete additional problems as well.
- Try to keep PSP least 80% no matter what. It shall really help you stay focused and we have seen in the past that people with $\geq 80\%$, do well in interviews.

2. Attendance

- Try to maintain at least **85% attendance** either through classes or by watching recordings.
- Though I will recommend you to come to classes regularly because otherwise it may create backlogs.
- So, I expect all of you to attend live classes and if for any reasons you are unable to, then please send me a message stating the reason.



Contest will be organised after Intermediate Module

- It will be for **1.5 hours** and will be conducted within class duration followed by 75-1: Contest Discussion (Instructor shall be discussing contest problems).
- If for any reason you are unable to pass the contest, then we shall be having re-attempts as well, therefore nothing to worry. (We will provide more info on re-attempts and the passing criteria moving forward. Most likely, there will be 4 questions out of which 3 needs to be solved)
- Contests are critical to retain what you have learnt and measuring where you need improvement. Please take contests seriously.

FAQs

- Notes will be uploaded after the class.
- Assignment will be unlocked after the class ends.
- There is no deadline for assignments.

DO NOT CREATE
BACKLOGS.

Lecture Days:-

Default

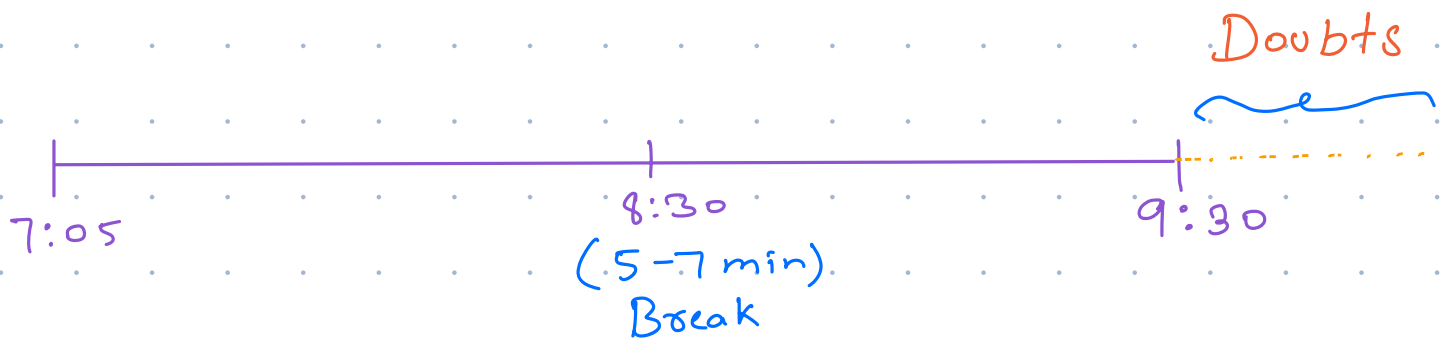
Monday

Wednesday

Friday

7:00 AM IST.

We start at 7:05 sharp



Doubts

Conceptual

↳ Harshal Gay.

Implementation

↳ Peers;

↳ T.A.;

↳ Chatbot.

Today's Agenda:-

1. Count of factors.
 2. Prime Numbers.
 3. Basic Maths.
 4. Iterations.
 5. Time Complexity (O).
-



Factor :-

If A is a factor of B ;
it means that A is completely divided
by $\underline{B}$.
 $B \div A = 0$

< **Question** >: Given N . Find the count of positive factors of N . ($N > 0$).

QUIZ

1. $N = 24$

1, 2, 3, 4, 6, 8, 12, 24

8.

QUIZ

2. $N = 10$

1, 2, 5, 10

4.

1 2 3 4 5 6 7 8 9 10

Count = $\cancel{0} \times \cancel{1} \times \cancel{2} \times \cancel{3} \times \cancel{4}$



< / > Pseudo Code

```

int CountFactors ( int N ) {
    int count = 0;
    for ( i = 1 ; i <= N ; i++ ) {
        if ( N % i == 0 ) {
            count ++;
        }
    }
    return count;
}

```

Assumption :- The machine on which this code is executed makes 10^8 iterations in 1 second.

N	Iterations	Execution Time
10^8	10^8	1 sec.
10^9	10^9	$\frac{1}{10^8} \times 10^9 = 10$ sec.
10^{18}	10^{18}	$\frac{1}{10^8} \times 10^{18} = 10^{10}$ sec.

≈ 300 years.



Optimisation If i is a factor of N ;
 then N/i is also a factor of N .

Dividend = Divisor $\times$ Quotient + Remainder.

$$N = 24$$

i	N/i
-----	-------

1	<	24
2	≤	12
3	≤	8
4	≤	6
6	>	4
8	>	3
12	>	2
24	>	1

$$N = 100$$

i	N/i
-----	-------

1	≤	100
2	≤	50
4	≤	25
5	≤	20
10	≤	10
20	>	5
25	>	4
50	>	2
100	>	1

Observation:-

1. The factors are repeating themselves.

2. Try start repeating after $i \leq N/i$; $i^2 \leq N$

Taking a square root on both sides

$$i \leq \sqrt{N}$$

Approach:-

Iterate until $\sqrt{N}$; If

i is a factor ; $count += 2$; Handle perfect square.



</> Pseudo Code

```
int countFactors (int N) {
```

```
    int count = 0; i i ≤ N
```

```
    for (i = 1; i ≤ sqrt(N); i++) {
```

```
        if (N % i == 0) {
```

```
            if (i == N/i) {
```

```
                count++;
```

```
            } else {
```

```
                count += 2;
```

```
            }
```

```
    }
```

```
    return count;
```

```
}
```

$$\frac{a^x}{a^y} = a^{x-y}$$

$$\frac{1}{10000}$$

N	Iterations	Execution Time
10^8	10^4	$\frac{1}{10^8} \times 10^4 =$
10^9	$10^4 \times \sqrt{10}$	$3.14 \times 10^4 =$
10^{18}	10^9	$\frac{1}{10^8} \times 10^9 =$

$$0.0001 \text{ sec}$$

$$0.00314 \text{ sec}$$

$$10 \text{ sec.}$$

300 years

→ 10 sec



Prime Numbers :- A no. having exactly 2 factors.

QUIZ

How many prime numbers are there?

10, 11, 23, 2, 25, 27, 31

Ans = 4.

< Question >: Given a number N. Check if it is prime or not.

< /> Pseudo Code

```
boolean checkPrime (int N) {  
    if (CountFactors (N) == 2) {  
        return true;  
    }  
    else {  
        return false;  
    }  
}
```



Range

$[a, b] \rightarrow$

$(a, b) \rightarrow$

$[a, b) \rightarrow$

$(a, b] \rightarrow$

QUIZ

$[3, 10] \rightarrow 3, 4, 5, 6, 7, 8, 9, 10$
 $= 8 \quad = 10 - 3 + 1$

QUIZ

$[a, b] \rightarrow \underline{b - a + 1}$

$[a, b) \rightarrow b - a + 1 - 1$

$(a, b] \rightarrow b - a$

$(a, b) \rightarrow b - a - 1$

Sum of first N numbers

QUIZ

$$S = 1 + 2 + 3 + 4 + \dots + 100$$

$$+ S = 100 + 99 + 98 + 97 + \dots + 1$$

$$2S = 101 \times 100$$

$$S = \frac{101 \times 100}{2} = 5050$$

Sum of first N natural numbers:

$$S = \frac{N \times (N+1)}{2}$$



What is a iteration?

No. of times a loop is getting executed.

Quiz- 7

How many times will the below loop run ?

```
for(i -> 1 to N)
{
    if(i == N) break;
}
```

$N=5$

$1 = 1 \times 2 \times 3 \times 4 \times 5$

N

Quiz- 8

How many iterations will be there in this loop ?

```
for(i -> 0 to 100){
    s = s + i + i^2;
}
```

$\rightarrow 100 - 0 + 1 = 101$

$[0, 100]$



Quiz- 9

How many iterations will be there in this loop?

```
func(){  
    for(i -> 1 to N){  
        if(i % 2 == 0){  
            print(i);  
        }  
    }  
    for(j -> 1 to M){  
        if(j % 2 == 0){  
            print(j);  
        }  
    }  
}
```

$N + M$

Arithmetic Progression

A series where difference b/w two consecutive terms is constant.

ex:- a) 2, 4, 6, 8, 10, ...

b) 3, 1, -1, -3, -5, ...

a = 1st term

d = Common difference

Nth term

H.W.

Sum of first
N terms

H.W.

8:50

Geometric

- Series where the ratio of two consecutive terms is same.

5, 10, 20, 40, 80, 160, 320

5

a = First term

r = Common ratio

$$5 \times 2^4 = 5 \times 16 = 80$$

Nth term

$$a \cdot r^{N-1}$$

Sum of first
N terms

$$\frac{a(r^N - 1)}{(r - 1)}$$

Comparing 2 algorithms:-

Ramesh
↓
15 sec
(Windows XP)

↓ macbook m2
7 sec (C++)

Suresh

↓
10 sec
(Macbook m3
(Python))

↓ C++
5 sec:

We can conclude that some E.T. depends on various factors, it is not a good parameter to evaluate 2 algorithms.

∴ The correct criteria to compare

2 algos should be the number of iterations.



Comparing Iterations of two algorithms using Graphs

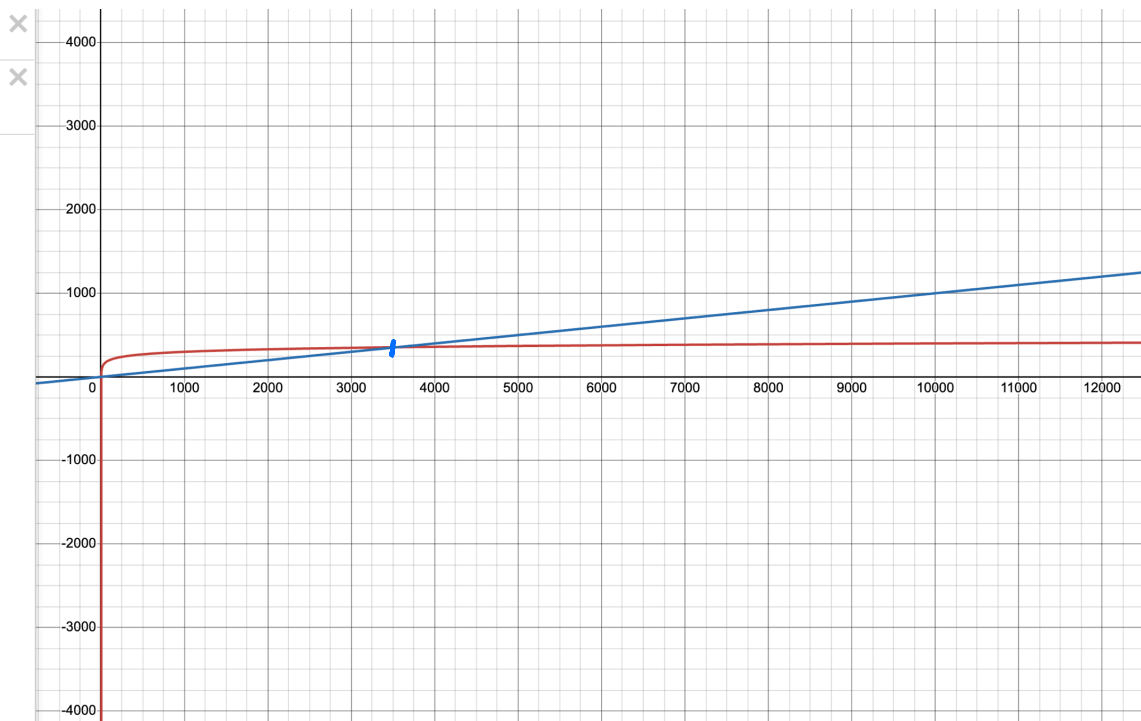
Winner. → Ramesh $O(\log N)$
 $100 \log N$

Suresh $O(N)$
 $N/10$

1
2
3

$$y = 100 \cdot \log N$$

$$y = \frac{N}{10}$$



On fn

$N < 3500$; Sures's algo is better.

but after

$N > 3500$; Rames's algo is better.

Asymptotic Analysis of Algorithms:-

or $B.T.O(O)$ means analysis also for larger input. ✓ To determine the T.C. of that algo:

Steps to calculate B.T.O.

1. Cal. the no. of iterations of the code.
2. Neglect the lower order term.
3. Ignore the constant coefficient.

Comparison Order:

$$\log(N) < \sqrt{N} < N < N \log(N) < N \sqrt{N} < N^2 < N^3 < 2^N < N! < N^N$$

Using an example

$$N = 36$$

$$5 < 6 < 36 < 36 \cdot 5 < 36 \cdot 6 < 36^2 < 36^3 < 2^{36} < 36! < 36^{36}$$

Examples:

$$\textcircled{1} \quad 4N^2 + 3N + 1 = \underline{\underline{O(N^2)}}$$

$$\textcircled{2} \quad 4N^2 + 3N + 6\sqrt{N} + 5N + 8 = \underline{\underline{O(N^2)}}$$

Ques 10:

$$4N + 3N \log(N) + 1 = \underline{\underline{O(N \log N)}}$$

$$\log_2 100 = 6.6..$$

Quiz 11 :

$$\sqrt{100} = 10$$

$$4N \log N + 3N \sqrt{N} + 10^6$$
$$\underline{O(N \sqrt{N})}$$

Why neglecting Lower Order Terms ?

$$\frac{1000}{11000} \approx 100$$

Let total iterations = $N^2 + 10N$

Contribution - $10N$ (Lower Order Term)

$$= \frac{10N}{N^2 + 10N} \approx 100$$

N	Total Iterations	Iterations due to L.O.T.	Percentage Contribution
10	200	100	50%
100	$10^4 + 10^3$	10^3	$\frac{10^3}{10^4 + 10^3} \approx 100\%$
10^4	$10^8 + 10^5$	10^5	$\frac{10^5}{10^8 + 10^5} = 0.1\%$

$\therefore$ We can conclude that contribution of L.O.T. will become negligible on increasing the value of N .

Why Neglecting Constant Coefficients 2.

Changing coefficients doesn't affect the graph in $\mathcal{O}(1)$ for large N . Thus, no change in input.

$$N=10;$$

$$1, 2, 5, 10$$

$$ans = 4.$$

$$1 \times 1 \leq 10$$

$$2 \times 2 \leq 10$$

$$3 \times 3 \leq 10$$

$$4 \times 4 \leq 10$$

$$1 \times 1 \leq 10$$

$$2 \times 2 \leq 10$$

$$3 \times 3 \leq 10$$

$$4 \times 4 \leq 10$$

$$ans = 4$$

Issues with Big(O) :

Issue 1: We cannot always say which algo is better for smaller values of N.

Algo 1:- $10^3 N \rightarrow O(N)$

Algo 2:- $N^2 \rightarrow O(N^2)$

N	Algo 1 ($10^3 N$)	Algo 2 (N^2)	Winner
100	10^5	10^4	<u>Algo 2.</u>
10^3	10^6	10^6	<u>Equally.</u>
$7 \cdot 10^3$			<u>Algo 1</u> is winner.

Issue 2:

Two algs can have same Big O but their actual iterations may vary & thus one may be better than the other.

Q:- Give N ; print all the odd no's.
for 1 to N .

N

$O(N)$

Code 1:-

```
for (int i = 1; i <= N; i++) {  
    if (i % 2 == 1) {  
        print(i);  
    }  
}
```

< 4.2

$N/2$
 $O(N)$

Code 2:-

```
for (int i = 1 ; i <= N ; i += 2) {  
    |  
    3  
    print (i);  
}
```

Although B.O of both C1 & C2 are same, Code 2 is definitely an optimal way to approach the problem.